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RESEARCH ARTICLE

Accurate and Cost-Efficient DLPNO-CCSD(T)-Based Feller-Peterson-Dixon Protocols Calibrated on Association Reactions of Li^+ Ion With Mixed Organic Carbonates

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ABSTRACT

We present a benchmark set LIMIXCARB_RE12 comprising 12 reference DLPNO-CCSD(T_1)/CPS(6,7)/CBS(cc-pwCVTZ/cc-pwCVQZ)//PBE0-D3(BJ)/def2-TZVP binding energies for the sizeable (up to 69 atoms) clusters of Li^+ ion with mixed cyclic and linear organic carbonates. A number of computationally cheaper DLPNO-CCSD(T)-based Feller-Peterson-Dixon protocols including contributions due to core-valence electron correlation, using more accurate iterative triples correction (T_1) and tighter-than-default PNO settings are examined with respect to their accuracy and efficiency. Particular splittings of the total binding energy into components allow maintaining high accuracy (deviations less than 0.2 kcal mol^{-1}) at significantly reduced computational cost. Much faster convergence of the DLPNO-CCSD(T) binding energies to the reference values is reached if Ahlrichs' def2 basis sets are used instead of their correlation-consistent Dunning counterparts. Evaluation of the DFT approximations against the LIMIXCARB_RE12 reveals double hybrid PWPB95-D4 in conjunction with CBS(def2-TZVPP/def2-QZVPP) extrapolation to be the best performer with mean signed deviation (MSD) of only −0.1 kcal mol^{-1} followed by r^{2}SCAN-D4/D3(BJ) and r^{2}SCAN-3c (MSD < 1 kcal mol^{-1}), while hybrid B3LYP and PBE0 functionals complemented with D3(BJ) or D4 dispersion corrections are clearly inferior. The obtained results provide a guide for the accurate calculations of the binding energies of the microsolvated clusters and can be used for the development and validation of the emerging computational methods.

1 | Introduction

The coupled cluster with single and double excitations complemented with perturbative triples correction (CCSD(T)) method [1] has been long recognized as the highly accurate approach to compute electron correlation energies in molecules and in turn, their individual properties as well as reaction energies [2]. It is often considered as a theory level of choice to generate reference data in benchmark studies [3]. However, almost prohibitive computational cost of the CCSD(T) method in case of medium-sized molecules and in conjunction with large basis sets required to ensure convergence of calculations is equally well known. The steep scaling of CCSD(T) can be reduced by exploiting the locality of electron correlation [4] giving rise to a number of local CCSD(T)-based approaches described elsewhere [5, 6]. Here we focus on the widely currently used domain-based local pair natural orbital (DLPNO)-CCSD(T) method [7–11] of Neese and co-workers. A number of studies reported encouraging performance of the DLPNO-CCSD(T) approach in reproducing of the experimental data [12–15] and

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canonical CCSD(T) results [16–19]. Moreover, this method itself has been extensively used [20–27] as a reference level of theory for systems which could not be treated within CCSD(T) due to their relatively large size. It should be noted, that despite its essentially black-box fashion, appropriate use of the DLPNO-CCSD(T) approximation requires careful selection of the related technical settings which can significantly influence the obtained results. First, the dimension of the PNO space is determined by T_{CutPNO} threshold, the PNOs with smaller occupation numbers are excluded from the correlation space. Tightening this threshold leads to a decrease of the error with respect to canonical CCSD(T) method which converges to zero at $T_{\mathrm{CutPNO}} = 0$, but at the cost of a dramatic increase in the computational time and memory requirements. Altun *et. al* found [28] exponential convergence of the DLPNO-CCSD(T) correlation energy with respect to T_{CutPNO} parameter and derived a simple two-point extrapolation procedure to the complete PNO space (CPS) limit. Further studies [29, 30] confirmed the fruitfulness of this approach and even suggested [31] it to be a standard practice in calculations of the transition metal spin-state energetics. An interesting alternative scheme to approach the limit of the complete PNO space involving MP2 correction has been also recently suggested [32]. The second important question to be addressed concerns accounting for the correlation of outer-core electrons with each other and with valence electrons. It becomes especially significant [33] in case of transition metal complexes and molecular clusters and requires careful definition of the frozen core (FC) [34, 35]. Finally, it is now well established from a variety of studies, [17, 23, 24, 31, 36–38] that the accuracy of the DLPNO-CCSD(T) calculations can be improved if the iterative triples (IT) correction [39, 40] (T_1) is employed instead of its original non-iterative (T_0) counterpart [10].

In principle, all the above-mentioned issues could be properly taken into account within either single or only a few calculations, such as DLPNO-CCSD(T_1) with *VeryTightPNO* settings [41] ($T_{\mathrm{CutPNO}} = 10^{-8}$, $T_{\mathrm{CutMKN}} = 10^{-4}$, $T_{\mathrm{CutPairs}} = 10^{-6}$) or CPS(6,7) extrapolation which requires two consecutive calculations with *TightPNO* settings and $T_{\mathrm{CutPNO}} = 10^{-6}$ and $T_{\mathrm{CutPNO}} = 10^{-7}$. Outer-core electrons should be included in the correlation treatment (all-electron, AE variant) and sufficiently large basis sets (at least of triple- and quadruple-ζ quality) containing weighted core functions [42] with the subsequent complete basis set (CBS) extrapolation should be employed. However, such calculations can be still prohibitively expensive for the molecules containing 50–100 atoms and further tuning of the computational protocol is required. Composite ab initio approaches [43] approximating the energy obtained with a high-level method as a sum of separate contributions calculated at lower and more economical levels of theory are viable and widely used workarounds to this problem. For example, well-established Gaussian-*n* (G*n*), [44] Weizmann-*n* (W*n*), [45] correlation consistent Composite Approach (ccCA) [46] and CBS [47] methodologies typically ensure [45] the accuracy of 1–10 kJ mol^{-1} and essentially represent some fixed and pre-defined series of consecutive calculations yielding a final energy. Some emerging composite approaches incorporate DLPNO-CCSD(T) approximation (DLPNO-ccCA [48]) or its explicitly correlated F12 version (as in L-W1X protocol [49]) to alleviate high computational demands of the canonical coupled cluster method.

Unlike the above-mentioned fixed-recipe model chemistries, flexible Feller-Peterson-Dixon (FPD) approach [50–52] allows fine tuning of the components of the composite protocol depending on the target property and desired accuracy. Used together with the DLPNO-CCSD(T) method, it enables separate evaluation of the contributions to the total molecular or reaction energy due to tighter-than-default PNO settings, core-valence (CV) correlation and iterative triples correction using smaller basis sets. Such estimation of the component(s) of the total DLPNO-CCSD(T) energy has been a rather common practice in our previous studies [53–55] and in the research of other groups [23, 56, 57]. At the same time, to the best of our knowledge, systematic comparison of the accuracy and efficiency of the different DLPNO-CCSD(T) based FPD schemes has not been performed to date. Our recent work [58] was limited to the relatively small H,C,O organic compounds and the question on the performance of FPD DLPNO-CCSD(T) approaches for the medium-sized species remains open.

Besides, the analysis of the recent literature reveals no consensus regarding the choice of basis sets to be used in conjunction with the DLPNO-CCSD(T) method. Both Ahlrichs' (def2) and Dunning basis sets are widely represented, in some cases augmented with diffuse functions. Ideally, larger Dunning basis sets (aug-)cc-p(wC)V*n*Z (n = T, Q) should be used to obtain most accurate energies at the CBS limit. However, in order to reduce computational demands their cheaper (ma-)def2-*n*ZVPP counterparts are also often considered. Comparing reaction energies and barrier heights of the enzymatically catalyzed reactions, Wappet and Goerigk found [59] almost similar accuracy of the CBS(aug-cc-pVTZ/aug-cc-pVQZ) and CBS(ma-def2-TZVPP/ma-def2-QZPP) values. The latter combination was also found to provide robust results for CHAL336 benchmark set [60] comprising reaction energies of the chalcogen-bonded dimers. Besides, the same def2 basis sets (without diffuse functions) were used in the recent studies [23, 24] by Wappet and Goerigk to obtain benchmark quality reaction energies and barrier heights for metalloenzyme reactions. Similar observations on the comparable accuracy of CBS(aug-cc-pVTZ/aug-cc-pVQZ) and CBS(def2-TZVPP/def2-QZVPP) reaction energies were reported [57] by Hansen and co-workers for an exemplary reaction from the ROST61 set.

In this work, we examine the accuracy and efficiency of various DLPNO-CCSD(T) based FPD approaches utilizing either Ahlrichs' or Dunning basis sets. The comparison is performed for the specially developed benchmark set LIMIXCARB_RE12 comprising 12 reference reaction energies at the highest feasible reference level of theory, DLPNO-CCSD(T_1)/CPS(6,7)/CBS(cc-pwCVTZ/cc-pwCVQZ). Model reactions of the type:

$$\mathrm{Li^+ + 2CC + 2LC = Li^+(CC)_2(LC)_2} \quad (1)$$

describe the microsolvation of Li^+ ion in the mixture of cyclic (CC) and linear (LC) organic carbonates. Note, that the considered clusters of Li^+ ion with mixed carbonates are sizable species containing up to 69 atoms and requiring more than 4000 basis functions for the highest-level calculations. Apart from the needs of the present study, the structures of the microsolvated $Li^+(CC)_2(LC)_2$ clusters and accurate binding energies are important for the further development of the lithium ion batteries

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(LIBs) [61, 62]. Currently, 50% CC—50% LC mixtures are often utilized in commercial lithium ion battery electrolytes [63]. As a byproduct, a number of contemporary density functional theory (DFT) approximations are examined against the LIMIXCARB_RE12 set.

2 | Design of the LIMIXCARB_RE12 Benchmark Set

The development of the Li-ion electrolytes has convincingly demonstrated that mixed cyclic and linear organic carbonate (OC) solvents should be used in order to impart the merits of the individual substances into the resultant mixture [64]. For this reason, and in continuation of our previous study [65], we selected 4 cyclic (CC) and 3 linear (LC) carbonates relevant [61, 66–68] to LIBs to compose 12 mixed carbonates of the type $Li^+(CC)_2(LC)_2$ (Figure 1). The following solvents utilized in the commercial Li- and Na-ion batteries were considered: propylene carbonate (PC), vinylene carbonate (VC), ethylene carbonate (EC), 1,2-butylene carbonate (BC), ethyl methyl carbonate (EMC), dimethyl carbonate (DMC) and diethyl carbonate (DEC).

Next, energies of the association reactions described by Equation (1) were computed using different computational protocols detailed in Section 3. Note, that the reactions of this type appear in the framework of the cluster-continuum approach [69, 70] and are used [71] to derive solvation free energies of ions.

The structures of the reactants and reference binding energies comprise the LIMIXCARB_RE12 benchmark set and are available from the GitHub repository of our group.

3 | Computational Details

3.1 | Conformational Search and Geometry Optimization

Clusters of Li^+ ion with mixed carbonates are expected to possess rich conformational landscapes arising from possible different arrangements of the carbonate molecules with respect to each other and central ion as well as from their inherent conformational flexibility (for BC, DMC, EMC and DEC). Recently, we introduced the Uniconf conformer generator [72] and extensively tested it on transition metal complexes, [73] molecular clusters [74] and cluster ions [65]. Here it was also used to generate the sets of 20 structurally diverse conformations for each of $Li^+(CC)_2(LC)_2$ clusters. The corresponding Uniconf input files are provided in the ESI. The following timings were obtained when running the code on the largest $Li^+(BC)_2(DEC)_2$ clusters: 7 min 25 s on a 6 core Intel i5-10600K 16 Gb RAM workstation and 4 min 7 s on a 16 core AMD Ryzen 9 5950X 64Gb RAM workstation.

The total of 240 generated structures were optimized using the PBE functional [75, 76] complemented with D3(BJ) dispersion correction [77, 78] and paired with def2-SVP basis set

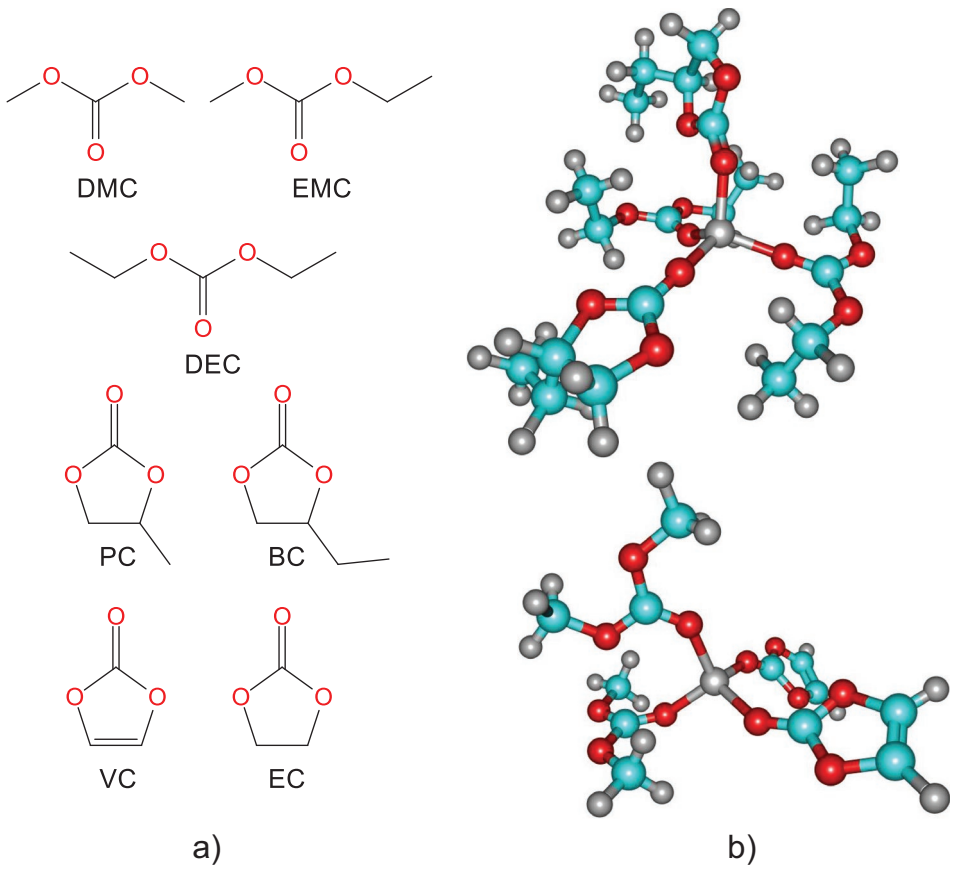


FIGURE 1 | Molecular structures of the individual organic carbonates (a), and spatial structures of several mixed carbonates ($Li^+(BC)_2(DEC)_2$) and $Li^+(VC)_2(DMC)_2$ (b). Atom color coding: Carbon (turquoise), oxygen (red), hydrogen (gray), and lithium (gray).

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[79] as implemented in the ORCA 5.0 program [80–84]. DFT exchange-correlation integration grid DEFGRID3 was specified. Substantially tighter-than-default geometry optimization stopping criteria were chosen: energy change $TolE = 1 \times 10^{-6}$ a.u., root-mean-squared (RMS) gradient $TolRMSG = 3 \times 10^{-5}$ a.u. and RMS maximum gradient $TolMaxG = 1 \times 10^{-4}$ a.u. Optimized structures were examined [85–87] for the presence of possible duplicates and in case of $Li^+(VC)_2(DMC)_2$ four similar structures were removed.

Single-point energies (SPEs) of the optimized structures were evaluated using the range-separated hybrid *meta*-GGA ωB97M-V functional [88] with VV10 dispersion correction [89] invoked [90] in the default post-SCF way. These calculations were performed with def2-TZVP and def2-QZVP basis sets [79] with a subsequent CBS extrapolation using the two-point $(1 + l_{max}/2)^{-4}$ scheme of Martin [91]. The conformation with the lowest ωB97M-V/CBS energy was selected for the further processing. Lowest energy conformers of BC, DMC, EMC and DEC were taken from Ref. [65].

Finally, the structures of the 12 selected $Li^+(CC)_2(LC)_2$ conformations and organic carbonates were re-optimized using hybrid PBE0 functional [92] in conjunction with D3(BJ) dispersion correction [77, 78] and def2-TZVP basis set [79]. All convergence tolerances were the same as for PBE-D3(BJ)/def2-SVP optimizations described above. Calculation of vibrational frequencies for the optimized structures revealed no imaginary modes and confirmed them to correspond to local energy minima. The obtained structures were then used for the DLPNO-CCSD(T) and DFT SPE evaluations.

3.2 | Single-Point Energy Evaluation

To evaluate the performance of different energy decomposition schemes via the Feller-Peterson-Dixon approach, 76 separate DLPNO-CCSD(T) calculations were performed for each of 12 clusters of Li^+ with mixed carbonates, corresponding solvent molecules and Li^+ ion. The following technical settings were independently varied:

- Frozen-core (FC) or all electron (AE) approximation;
- Semi-canonical (T_0) or iterative (T_1) triples correction;
- T_{CutPNO} cut-off;
- Basis set.

The default ORCA 6.0 FC approximation implies that 1s core electrons of atoms C and O are "frozen", while in the AE approach these electrons are included in the post-HF correlation treatment. For T_{CutPNO} the following different values were specified: 1.0×10^{-5}, 1.0×10^{-6} and 1.0×10^{-7} (corresponds to *TightPNO* settings). In addition, the default *NormalPNO* settings (with $T_{CutPNO} = 3.33 \times 10^{-7}$) were applied. The energies with different T_{CutPNO} thresholds were used for the complete PNO space (CPS) extrapolation given [28] by the formula:

$$E_{corr(CPS)} = E_{corr(X)} + F\left(E_{corr(Y)} - E_{corr(X)}\right) \quad (2)$$

where $E_{corr(X)}$ is the correlation energy obtained with $T_{CutPNO} = 10^{-X}$ and $Y = X + 1$. The F value of 1.5 was used for both 5/6 and 6/7 extrapolations as recommended by Altun et al. [28].

In all DLPNO-CCSD(T) calculations the default T_{CutMKN} cut-off parameter of 1.0×10^{-3} was tightened to 1.0×10^{-4} as it was found [18] to correct energies of some Diels-Alder reactions strongly deviating from their reference CCSD(T) counterparts. "*UseFullLmp2Guess True*" option was also applied to increase the accuracy of the calculations.

SPE calculations were performed with Dunning and Ahlrichs' families of basis sets. In the first case cc-pwCV*n*Z (n = D, T, Q) basis sets [93] were assigned to Li atom, while cc-pV*n*Z basis sets [94] were employed on atoms C and O in case of FC calculations and their weighted core-valence cc-pwCV*n*Z analogues [42] – within the AE approximation. Hydrogen atom was described using cc-pV*n*Z basis sets [94]. Throughout this paper, these combinations of basis sets will be referred to as cc-p(wC)V*n*Z in case of FC calculations and cc-pwCV*n*Z—for AE calculations. Auxiliary correlation fitting basis sets (cc-p(wC)V*n*Z/C) [95, 96] were employed as requested in the resolution-of-the-identity (RI) approximation. For Li atom these /C basis sets were automatically generated using the *AutoAux* routine [97].

To avoid repeating HF SPE calculations, all of them were performed prior to the target DLPNO-CCSD(T) energy evaluations with the matching basis sets and orbitals were read via the *MOREAD* option.

Calculations employing Ahlrichs' def2-*n*ZVPP (n = T, Q) basis sets [79] were performed within the FC approximation. Corresponding def2-*n*ZVPP/C [98] basis sets were utilized.

3.3 | Components of Reaction Energies Accounted in the Feller-Peterson-Dixon Approach

The energies of the Equation (1) reactions were obtained using different forms of the composite FPD approach:

$$\Delta_r E = \Delta_r E_{CBS} + \Delta_r E_{CV} + \Delta_r E_{IT} + \Delta_r E_{PNO} \quad (3)$$

where $\Delta_r E_{CBS}$ is the reaction energy at the complete basis set (CBS) limit, $\Delta_r E_{CV}$ is the contribution due to core-valence electron correlation, $\Delta_r E_{IT}$ is the impact of using iterative triples correction (T_1) instead of its semi-canonical T_0 counterpart and $\Delta_r E_{PNO}$ quantifies the influence of using tighter-than-default PNO settings on the reaction energy.

Only the first term in the Equation (3) is mandatory and appears in all the examined FPD schemes. The $\Delta_r E_{CBS}$ is evaluated using the energies of the reactants extrapolated to CBS using well-tested [58, 99] two-point $(1 + l_{max}/2)^{-4}$ scheme of Martin [91]. In all cases extrapolation was performed for triple- (TZ) and quadruple-ζ (QZ) basis set pairs. An alternative separate CBS extrapolation [100] of the HF [101, 102] and correlation [103] energies.

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$$E_{\mathrm{SCF}}^{(\infty)} = \frac{E_{\mathrm{SCF}}^{(X)} \cdot \exp\left(-\alpha\sqrt{Y}\right) - E_{\mathrm{SCF}}^{(Y)} \cdot \exp\left(-\alpha\sqrt{X}\right)}{\exp\left(-\alpha\sqrt{Y}\right) - \exp\left(-\alpha\sqrt{X}\right)} \tag{4}$$

$$E_{\mathrm{corr}}^{(\infty)} = \frac{X^{\beta} \cdot E_{\mathrm{corr}}^{(X)} - Y^{\beta} \cdot E_{\mathrm{corr}}^{(Y)}}{X^{\beta} - Y^{\beta}} \tag{5}$$

which is currently the default scheme in ORCA program is also considered. In Equations (4) and (5), $X=3$ and $Y=4$ are successive cardinal numbers of basis sets, α and β are basis set specific constants. Their optimized values for def2-*n*ZVPP ($\alpha_{34}=7.88$, $\beta_{34}=2.97$) and cc-p(wC)V*n*Z ($\alpha_{34}=5.46$, $\beta_{34}=3.05$) basis set families were retrieved from the Ref. [100]

The contribution of the core-valence correlation, $\Delta_r E_{\mathrm{CV}}$, was computed in those FPD schemes utilizing Dunning basis sets, where the dominant $\Delta_r E_{\mathrm{CBS}}$ term was obtained in the FC approximation. It was estimated at the triple-ζ basis set as the difference between AE and FC DLPNO-CCSD(T_0)/*NormalPNO*/cc-p(wC)VTZ reaction energies. Recently we observed [58] an improved accuracy at nearly the same computational cost if $\Delta_r E_{\mathrm{CV}}$ effect is computed from the CBS-extrapolated individual E_{CV}(DLPNO-CCSD(T_0)/*NormalPNO*/cc-p(wC)V*n*Z) values of the reactants (n = D, T) and tested this scheme in this work. In order to directly evaluate the accuracy of the TZ and CBS extrapolated $\Delta_r E_{\mathrm{CV}}$ terms, we also obtained the best feasible evaluation of the CV effect as the difference between AE and FC DLPNO-CCSD(T_1)/CPS(6,7)/CBS(cc-pwCVTZ/cc-pwCVQZ) reaction energies.

For the FPD schemes containing $\Delta_r E_{\mathrm{CBS}}$ term obtained with more efficient semi-canonical triples correction (T_0), the $\Delta_r E_{\mathrm{IT}}$ effect was estimated as the difference between FC DLPNO-CCSD(T_1) and DLPNO-CCSD(T_0) reaction energies computed with triple-ζ basis sets. Reference $\Delta_r E_{\mathrm{IT}}$ values were obtained as the difference between AE DLPNO-CCSD(T_1)/CPS(6,7)/CBS(cc-pwCVTZ/cc-pwCVQZ) and DLPNO-CCSD(T_0)/CPS(6,7)/CBS(cc-pwCVTZ/cc-pwCVQZ) reaction energies.

Finally, if $\Delta_r E_{\mathrm{CBS}}$ contribution is obtained with the default *NormalPNO* settings, the impact of the extension of the PNO space is also estimated using triple-ζ basis sets as the difference between FC DLPNO-CCSD(T_0) reaction energies with "*Tight*" and *NormalPNO* settings. Three variants of the "*Tight*" settings were tested including those available via the corresponding *TightPNO* keyword, its estimated and more efficient CPS(5,6) substitute and presumably most accurate CPS(6,7) evaluation. The best evaluation of the PNO effect was obtained as the difference between AE DLPNO-CCSD(T_1)/CPS(6,7)/CBS(cc-pwCVTZ/cc-pwCVQZ) and DLPNO-CCSD(T_1)/*NormalPNO*/CBS(cc-pwCVTZ/cc-pwCVQZ) reaction energies.

In addition to the various DLPNO-CCSD(T) based FPD schemes, a number of DFT approximations were examined against LIMIXCARB_RE12 set including *meta*-generalized-gradient approximation (*m*GGA) r^{2}SCAN [104, 105] and hybrid B3LYP [106–108] and PBE0 [92] functionals complemented with D3(BJ) [77, 78] or D4 [109, 110] dispersion correction, hybrid M06 [111] with D3(0) [112] correction, range-separated hybrid *m*GGA ωB97M-V [88] with VV10 dispersion correction [89] invoked [90] in the default post-SCF way, composite B97-3c [113] and r^{2}SCAN-3c [114] and double hybrids (DHs) B2PLYP [115] and PWPB95 [116] paired with D4 dispersion correction. DFT calculations (except for the composite approaches) were performed with def2-TZVPP and def2-QZVPP basis sets with the subsequent CBS extrapolation utilizing either $(1+l_{\max}/2)^{-4}$ or Equation (4) scheme.

4 | Results and Discussion

This section has been divided into three parts. First, the accuracy of the individual components of the FPD approach (Equation 3) is evaluated against their best estimates. Second, reaction energies computed using different FPD protocols and corresponding relative timings are compared. Third, we discuss the performance of the DFT methods against the developed LIMIXCARB_RE12 benchmark set.

4.1 | Individual Components of the Feller-Peterson-Dixon Schemes

According to the Feller-Peterson-Dixon approach, the total reaction energy can be expressed as a sum of separate relevant contributions evaluated at the levels of theory which ensure the target accuracy at the reduced overall computational cost. In this work, we consider FPD schemes with up to 4 components (Equation 3). Two of them, $\Delta_r E_{\mathrm{CBS}}$ and $\Delta_r E_{\mathrm{CV}}$ are inherited from the original formulation of the FPD method, [50] and two remaining terms, $\Delta_r E_{\mathrm{IT}}$ and $\Delta_r E_{\mathrm{PNO}}$ are specific for the DLPNO-CCSD(T) method. In this section, we will compare three smaller contributions obtained at different levels of theory (Figures 2–4) with their best estimates (Section 3.3), while the dominant $\Delta_r E_{\mathrm{CBS}}$ terms are compared in the Section 4.2.

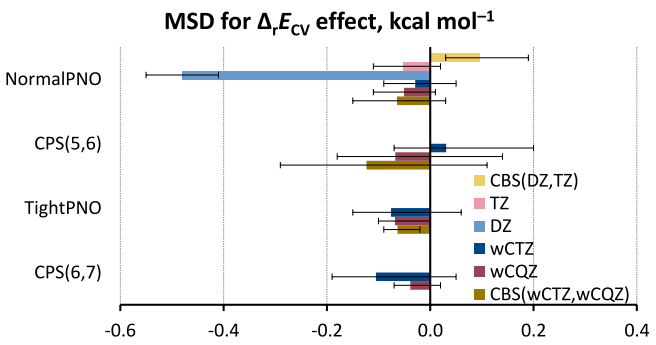


FIGURE 2 | Mean signed deviation (MSD) for the core—valence electron correlation (unfrozen 1s^2 electrons on C and O atoms) effect on the reaction energies. The $\Delta_r E_{\mathrm{CV}}$ effect for each reaction is computed as the difference between all-electron (AE) and frozen core (FC) DLPNO-CCSD(T_0/T_1) reaction energies and different PNO settings and basis sets. Reference $\Delta_r E_{\mathrm{CV}}$ were obtained at DLPNO-CCSD(T_1)/CPS(6,7)/CBS(cc-pwCVTZ/cc-pwCVQZ) level. Endpoints of each line indicate minimum and maximum deviations among the reactions. The following abbreviations of the basis sets are used: *n*Z = cc-p(wC)V*n*Z, wC*n*Z = cc-pwCV*n*Z (n = D, T, Q). In the case of CBS(DZ,TZ) variant individual E_{CV} contributions of reactants were extrapolated, see Section 3.3 for details.

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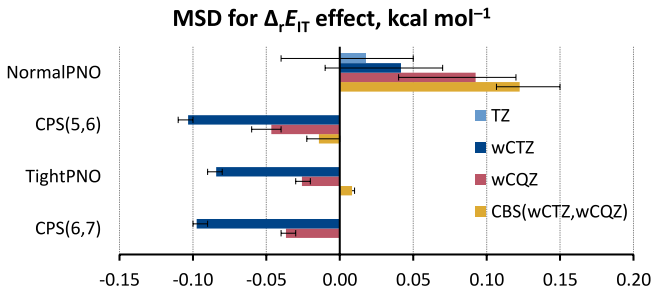


FIGURE 3 | Mean signed deviation (MSD) for the impact of the iterative triples (T_1) correction on the reaction energies. The $\Delta_r E_{IT}$ effect for each reaction is approximated as the difference between DLPNO-CCSD(T_1) and DLPNO-CCSD(T_0) reaction energies obtained with different PNO settings and basis sets. Reference $\Delta_r E_{IT}$ were obtained at AE-DLPNO-CCSD(*T*)/CPS(6,7)/CBS(cc-pwCVTZ/cc-pwCVQZ) level. Endpoints of each line indicate the minimum and maximum deviations among the reactions. The following abbreviations of the basis sets are used: *n*Z = cc-p(wC)V*n*Z, wC*n*Z = cc-pwCV*n*Z (*n* = D, T, Q).

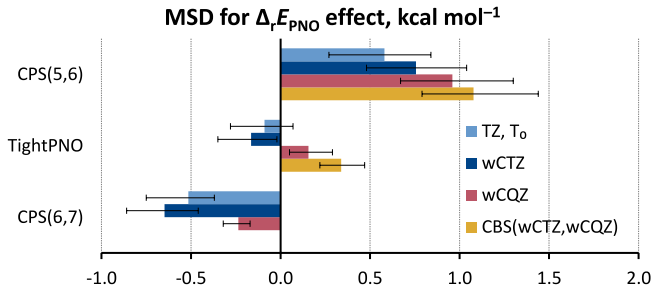


FIGURE 4 | Mean signed deviation (MSD) for the effect of PNO settings on the reaction energies. The $\Delta_r E_{PNO}$ effect for each reaction is calculated as the difference between CPS(5,6)/*TightPNO*/CPS(6,7) and *NormalPNO* DLPNO-CCSD(T_1) reaction energies obtained with different basis sets (or with semi-canonical T_0 triples in case of TZ basis set). Reference $\Delta_r E_{PNO}$ were obtained at AE-DLPNO-CCSD(*T*)/CPS(6,7)/CBS(cc-pwCVTZ/cc-pwCVQZ) level. Endpoints of each line indicate minimum and maximum deviations among the reactions. The following abbreviations of the basis sets are used: *n*Z = cc-p(wC)V*n*Z, wC*n*Z = cc-pwCV*n*Z (*n* = D, T, Q).

Average and maximum reference $\Delta_r E_{CV}$ (0.2/0.3 kcal mol^{-1}) and $\Delta_r E_{IT}$ (0.1/0.2 kcal mol^{-1}) values for the considered 12 reactions reveal small influence of the core-valence electron correlation and iterative triples correction on the reaction energies. As was pointed out in Section 3.2, the $\Delta_r E_{CV}$ term does not involve the effect of the CV correlation on Li atom as its $1s^2$ core electrons are included in the correlation procedure within the default ORCA frozen core settings. To quantify the performance of the cheaper alternative approaches for the evaluation of $\Delta_r E_{CV}$ and $\Delta_r E_{IT}$, the mean signed deviations (MSDs) were computed with respect to the reference values (Figures 2 and 3). Analysis of the Figure 2 shows that in most cases the average deviations are smaller than 0.1 kcal mol^{-1} with the only striking exception of the cc-p(wC)VDZ (abbreviated as DZ) level of theory. The $\Delta_r E_{CV}$ computed with the double-ζ basis sets are systematically and largely underestimated which corroborates the earlier observation [50] that at least triple-ζ sets should be used for the estimation of the CV effect. It should be also noted that the standard cc-p(wC)VTZ (denoted as TZ) basis set combination ensures quite small deviations (mean: −0.05, max: −0.11 kcal mol^{-1}) from the reference results. CBS(DZ, TZ) extrapolated $\Delta_r E_{CV}$ are slightly overestimated and are inferior to their TZ counterparts.

The impact of the substitution of the semi-canonical T_0 triples correction with the improved iterative (T_1) correction is almost negligible and does practically not depend on the basis set and PNO settings (Figure 3). We only note that cc-p(wC)VTZ (TZ) basis set seems to be an optimal choice due to its computational efficiency and very small deviations with respect to the reference $\Delta_r E_{IT}$ values. Similar results were obtained with def2 basis sets (Figure S1).

Corrections due to tighter-than-default PNO settings are much more sensible to the specific variant of the "*Tight*" settings (*TightPNO*, CPS(5,6) or CPS(6,7)) and basis set. Figure 4 demonstrates significant (1.2 kcal mol^{-1} on average) overestimation of $\Delta_r E_{PNO}$ in case of CPS(5,6) extrapolation. In contrast, the evaluation of $\Delta_r E_{PNO}$ with cc-p(wC)VTZ basis set, T_0 triples correction and CPS(6,7) settings which might be considered as a good compromise between the desired accuracy and computational demands yields underestimated values. Surprisingly, the lowest absolute MSD was found for the even more efficient combination of cc-p(wC)VTZ basis set, T_0 semi-canonical triples and *TightPNO* settings which does not require an additional DLPNO-CCSD(T) computation with $T_{CutPNO} = 10^{-6}$. Once again, similar conclusions can be drawn from the comparison of the $\Delta_r E_{PNO}$ obtained with def2 basis sets (Figure S2). The following section moves on to consider different combinations of $\Delta_r E_{CV}$, $\Delta_r E_{IT}$ and $\Delta_r E_{PNO}$ corrections with the $\Delta_r E_{CBS}$ terms in the framework of the FPD approach and compares the resulting reaction energies.

4.2 | Binding Energies According to the Different FPD Approaches

In this section the target reaction energies obtained with different FPD schemes are compared to the presumably most accurate AE DLPNO-CCSD(T_1)/CPS(6,7)/CBS(cc-pwCVTZ/cc-pwCVQZ) results (Figures 5–7). As indicated previously, the first term of Equation (3), $\Delta_r E_{CBS}$, makes the dominant (at least 98.7%) contribution to the total $\Delta_r E$, while each of the other effects are several orders of magnitude smaller. To explore the influence of the CBS extrapolation, we also evaluated this first term with TZ and QZ basis sets. It is obvious that the limiting step of a reaction energy evaluation are HF and subsequent DLPNO-CCSD(T) calculations for a $Li^+(CC)_2(LC)_2$ cluster. To approximately estimate computational efficiency of the computational protocols, we also provide relative timings required for these calculations on the example of $Li^+(EC)_2(EMC)_2$ cluster.

Figure 5 collects FPD schemes with $\Delta_r E_{CBS}$ term as well as the corresponding TZ and QZ variants computed using Dunning basis sets and default *NormalPNO* settings. It is immediately clear that the most computationally efficient TZ basis sets are not sufficient to obtain accurate reaction energies and produce underestimated (by 6.4 kcal mol^{-1} on average) values compared to the reference $\Delta_r E$. Basis sets of QZ quality expectedly yield better (average MSD = −2.4 kcal mol^{-1}), but still not accurate enough results. A remarkable improvement at a very small

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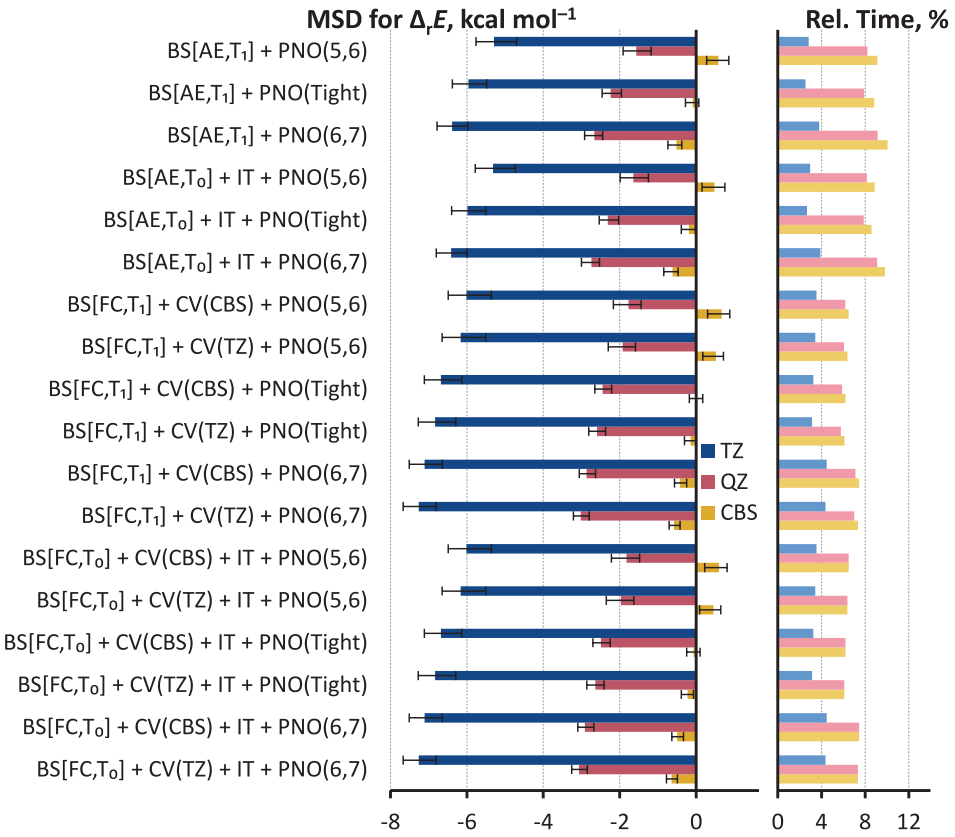


FIGURE 5 | Mean signed deviations (MSD) for the reaction energies obtained with different Feller-Peterson-Dixon (FPD) strategies for all clusters and relative times required for the corresponding calculations in case of $Li^+(EC)_2(EMC)_2$ cluster. Reference $\Delta_r E$ are computed at AE-DLPNO-CCSD(T_1)/CPS(6,7)/CBS(cc-pwCVTZ/cc-pwCVQZ) level of theory taken the most amount of time (100%). Endpoints of each line indicate minimum and maximum deviations among the reactions. The following abbreviations are used for the components of the FPD scheme: BS[FC, $T_{0/1}$] = $\Delta_r E$(FC-DLPNO-CCSD($T_{0/1}$)/*NormalPNO*/CBS(cc-p(wC)VTZ/cc-p(wC)VQZ)), BS[AE, $T_{0/1}$] = $\Delta_r E$(AE-DLPNO-CCSD($T_{0/1}$)/*NormalPNO*/CBS(cc-pwCVTZ/cc-pwCVQZ)), CV(TZ) = $\Delta_r E_{CV}$(DLPNO-CCSD(T_0)/*NormalPNO*/cc-p(wC)VTZ), CV(CBS) = $\Delta_r E_{CV}$ effect computed from CBS-extrapolated individual E_{CV}(DLPNO-CCSD(T_0)/*NormalPNO*/cc-p(wC)V*n*Z) values of the reactants (*n* = D, T), IT = $\Delta_r E_{IT}$(FC-DLPNO-CCSD(T)/*NormalPNO*/cc-p(wC)VTZ), PNO(Tight/(5,6)/(6,7)) = $\Delta_r E_{PNO}$(FC-DLPNO-CCSD(T_0)/(*TightPNO*/CPS(5,6)/CPS(6,7))/cc-p(wC)VTZ).

additional computational cost is achieved with CBS extrapolation. Maximum deviation from the reference $\Delta_r E$ is only 0.7 kcal mol^{-1}. Small MSD of −0.2 kcal mol^{-1} was obtained for the most computationally efficient FPD scheme combining $\Delta_r E_{CBS}$ derived using FC approximation and T_0 triples correction, and three other contributions calculated with triple-ζ basis sets (denoted as BS[FC,T_0] + CV(TZ) + IT + PNO(Tight) in Figure 5).

Accurate results although at a higher cost can be obtained if the core $\Delta_r E_{CBS}$ are computed with tighter-than-default PNO settings (Figure 6), except for the CPS(5,6) variant. For example, the most efficient CBS (FC, T_0, CPS(5,6)) + CV(TZ) + IT combination of this group yields MSD of 0.9 kcal mol^{-1}. Worse performance of the CPS(5,6) extrapolation stems from an unexpected dependence of the correlation energy on the T_{CutPNO} parameter obtained for the considered $Li^+(CC)_2(LC)_2$ clusters and individual LC and CC molecules if cc-p(wC)VQZ or cc-pwCVQZ basis sets are employed (see Supporting Information). In cases where FC approximation is used, the correlation energies with $T_{CutPNO} = 10^{-5}$ are the lowest ones and the expected smooth convergence with tightening of the T_{CutPNO} threshold is not observed. For this reason, we consider the CPS(5,6) $\Delta_r E_{CBS}$ to be unreliable and emphasize the need of the careful inspection of the dependence of the correlation energy on this truncation parameter before applying the CPS procedure. It is important to note that the smooth convergence is restored if the AE approximation is used for the DLPNO-CCSD(T_0/T_1) calculations in conjunction with the cc-pwCVQZ basis sets and the reference AE DLPNO-CCSD(T_1)/CPS(6,7)/CBS(cc-pwCVTZ/cc-pwCVQZ) reaction energies are not compromised. Other "*Tight*" settings including *TightPNO* and CPS(6,7) options ensure accurate (deviations are less than 0.5 kcal mol^{-1}) reaction energies, but are inferior to their *NormalPNO* counterparts (Figure 5) due their higher computational demands.

Finally, the performance of the Ahlrichs' def2 basis sets in conjunction with DLPNO-CCSD(T) method will be discussed. Comparison of Figure 7 with Figures 5 and 6 reveals two main differences between the considered basis set families. First, in case of def2 family the reaction energies converge much faster to the reference values with an increase of the basis set size (TZ → QZ → CBS). Second, in many cases def2-TZVPP basis set already ensures quite robust $\Delta_r E$ values with the absolute deviations below 0.5 kcal mol^{-1}. For example, FC DLPNO-CCSD(T_1)/CPS(6,7)/def2-TZVPP calculation yields MSD = 0.1 kcal mol^{-1} and can be considered as a cheap substitute of the expensive reference approximation. Interestingly, good performance of the def2-TZVP basis set in conjunction with LPNO-CCSD method

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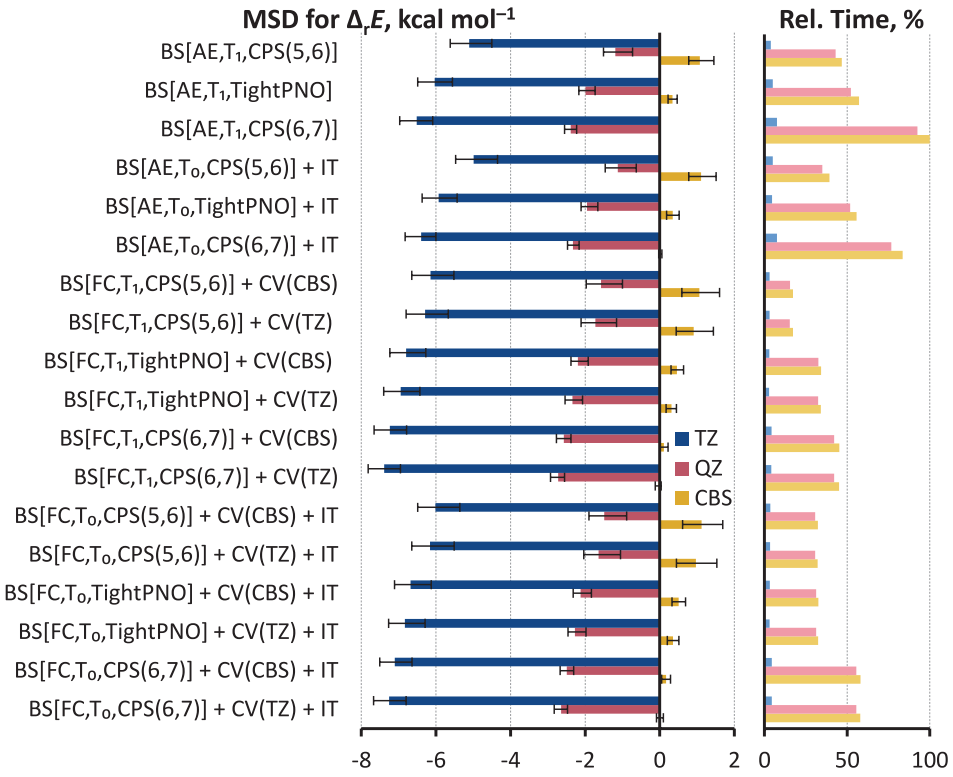


FIGURE 6 | Mean signed deviations (MSD) for the reaction energies obtained with different Feller-Peterson-Dixon (FPD) strategies for all clusters and relative times required for the corresponding calculations in case of $Li^+(EC)_2(EMC)_2$ cluster. Reference $\Delta_r E$ are computed at AE-DLPNO-CCSD(T_1)/CPS(6,7)/CBS(cc-pwCVTZ/cc-pwCVQZ) level of theory taken the most amount of time (100%). Endpoints of each line indicate the minimum and maximum deviations among the reactions. The following abbreviations are used for the components of the FPD scheme: BS[FC,$T_{0/1}$,CPS(5,6)/CPS(6,7)/TightPNO] = $\Delta_r E$(FC-DLPNO-CCSD($T_{0/1}$)/(CPS(5,6)/CPS(6,7)/*TightPNO*)/CBS(cc-p(wC)VTZ/cc-p(wC)VQZ)), BS[AE, $T_{0/1}$, CPS(5,6)/CPS(6,7)/TightPNO] = $\Delta_r E$(AE-DLPNO-CCSD($T_{0/1}$)/(CPS(5,6)/CPS(6,7)/*TightPNO*)/CBS(cc-pwCVTZ/cc-pwCVQZ)), CV(TZ) = $\Delta_r E_{CV}$(DLPNO-CCSD(T_0)/*NormalPNO*/cc-p(wC)VTZ), CV(CBS) = $\Delta_r E_{CV}$ effect computed from CBS-extrapolated individual E_{CV}(DLPNO-CCSD(T_0)/*NormalPNO*/cc-p(wC)V*n*Z) values of the reactants (*n* = D, T), IT = $\Delta_r E_{IT}$(FC-DLPNO-CCSD(T)/*NormalPNO*/cc-p(wC)VTZ).

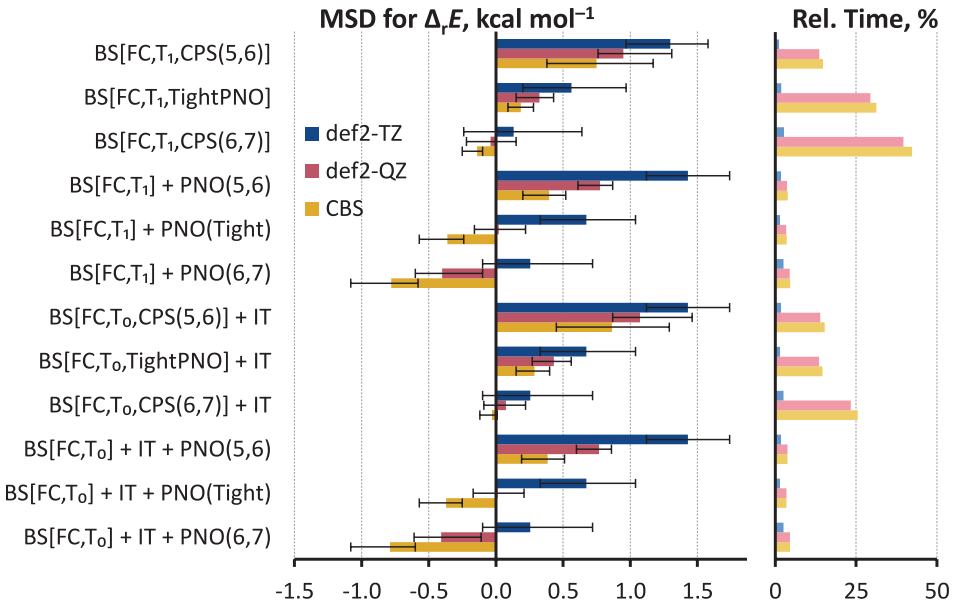


FIGURE 7 | Mean signed deviations (MSD) for the reaction energies obtained with different Feller-Peterson-Dixon (FPD) strategies for all clusters and relative times required for the corresponding calculations in case of $Li^+(EC)_2(EMC)_2$ cluster. Reference $\Delta_r E$ are computed at AE-DLPNO-CCSD(T_1)/CPS(6,7)/CBS(cc-pwCVTZ/cc-pwCVQZ) level of theory taken the most amount of time (100%). Endpoints of each line indicate minimum and maximum deviations among the reactions. The following abbreviations are used for the components of the FPD scheme: BS[FC,$T_{0/1}$,CPS(5,6)/CPS(6,7)/TightPNO] = $\Delta_r E$(FC-DLPNO-CCSD($T_{0/1}$)/(CPS(5,6)/CPS(6,7)/*TightPNO*)/CBS(def2-TZVPP/def2-QZVPP)), IT = $\Delta_r E_{IT}$(FC-DLPNO-CCSD(T)/*NormalPNO*/def2-TZVPP), PNO(Tight/(5,6)/(6,7)) = $\Delta_r E_{PNO}$(FC-DLPNO-CSD(T_0)/(*TightPNO*/CPS(5,6)/CPS(6,7))def2-TZVPP).

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[7, 8] eliminating the need of the CBS extrapolation was earlier reported by Anoop et al. [117].

All CBS-extrapolated energies discussed above were obtained using Martin $(1+l_{max}/2)^{-4}$ formula [91] for both HF and correlation energies. If instead the separate extrapolation [100] (Equations 4 and 5) of these two quantities is employed, the binding energies with Dunning and Ahlrichs' basis sets systematically decrease by ca. 0.4 and 0.2 kcal mol^{-1}, respectively. The absence of the accurate energies with larger basis sets does not currently allow recommending either of these two extrapolation schemes.

4.3 | Performance of the DFT Approximations for the Binding Energies

In this section the reference AE DLPNO-CCSD(T_1)/CPS(6,7)/CBS(cc-pwCVTZ/cc-pwCVQZ) binding energies are used to calibrate DFT approximations which are at least an order of magnitude faster compared to the cheapest DLPNO-CCSD(T) calculations (Figure 8). All DFT approaches except for r^{2}SCAN-D4 demonstrate the expected decrease of the MSD with the extension of the basis set (TZ → QZ → CBS). In case of PBE0 and B3LYP functional better statistical results were obtained with D4 dispersion correction compared to its D3(BJ) predecessor. However, we note that these popular approaches along with ωB97M-V and B2PLYP-D4 and composite B97-3c does not provide robust binding energies, the deviations from the reference values are more than 1.4 kcal mol^{-1}. Lower MSDs were obtained for M06-D3(0) functional, however the overall quality of the predictions is nonhomogeneous as indicated by the largest error bars (Figure 8). The absolute deviations below 1 kcal mol^{-1} were found for r^{2}SCAN functional with D3(BJ) and D4 dispersion corrections and also for the robust "Swiss army knife" composite r^{2}SCAN-3c approach. Taking into account the error bars indicating largest deviations and convergence behavior, we want particularly note good performance of the double hybrid PWPB95-D4 method which is in fact on par with the FPD DLPNO-CCSD(T) approximations considered in the previous section, especially if the CBS(TZ/QZ) extrapolation is applied. It can be recommended for the fast and accurate evaluations of the binding energies at least for the systems similar to the $Li^+(CC)_2(LC)_2$ clusters considered in this work.

5 | Conclusions

We introduce LIMIXCARB_RE12 set comprising 12 accurate AE DLPNO-CCSD(T_1)/CPS(6,7)/CBS(cc-pwCVTZ/cc-pwCVQZ)//PBE0-D3(BJ)/def2-TZVP binding energies for the clusters of Li^+ ion with mixed cyclic and linear organic carbonates. Using these values as reliable reference data, we perform a comprehensive evaluation of various Feller-Peterson-Dixon (FPD) protocols based on the DLPNO-CCSD(T) method. In addition, a number of contemporary and well-established DFT approaches are validated against LIMIXCARB_RE12 set. The following main conclusions can be drawn from the current study:

- Among three considered separate contributions to the binding energies due to accounting for the core-valence electron correlation ($\Delta_r E_{CV}$), employing more accurate iterative triples correction T_1 instead of its semi-canonical T_0 counterpart ($\Delta_r E_{IT}$) and using tighter-than-default PNO settings ($\Delta_r E_{PNO}$), the latter one is the most important.
- All these contributions can be efficiently evaluated using TZ basis sets.
- A caution should be exercised if extrapolation to the complete PNO space (CPS) is performed in case of the default

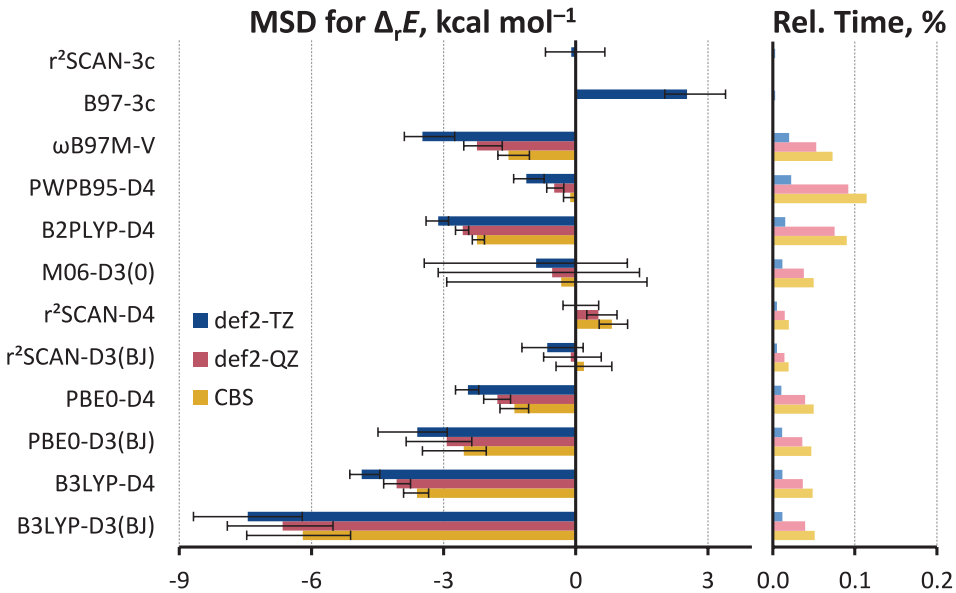


FIGURE 8 | Mean signed deviations (MSD) for the reaction energies obtained with different Feller-Peterson-Dixon (FPD) strategies for all clusters and relative times required for the corresponding calculations in case of $Li^+(EC)_2(EMC)_2$ cluster. Reference $\Delta_r E$ are computed at AE-DLPNO-CCSD(T_1)/CPS(6,7)/CBS(cc-pwCVTZ/cc-pwCVQZ) level of theory taken the most amount of time (100%). Endpoints of each line indicate minimum and maximum deviations among the reactions. The following abbreviations of the basis sets are used: Def2-*n*Z = def2-*n*ZVPP (*n* = T, Q). In case of the composite B97-3c and r^{2}SCAN-3c approaches their default modified def2-mTZVP and def2-mTZVPP basis sets were employed.

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frozen core (FC) approximation and in conjunction with QZ basis sets. For this FC/QZ combination we observed an unexpected non-smooth convergence of the correlation energy with respect to the T_{CutPNO} threshold which hampers CPS(5,6) extrapolation.

- Properly designed FPD approaches significantly reduce computational demands while preserving sufficient accuracy. For example, splitting of the total binding energy into $\Delta_r E_{CBS}$ term evaluated within the FC approximation and utilizing less expensive T_0 triples correction and $\Delta_r E_{CV}$, $\Delta_r E_{IT}$ and $\Delta_r E_{PNO}$ components computed with TZ basis sets yields mean signed deviation (MSD) of only $-0.2\,kcal\,mol^{-1}$ with respect to the reference values and requires (roughly) only 6.1% amount of the reference computational time.
- Ahlrichs' def2 basis sets in conjunction with DLPNO-CCSD(T) method ensure much faster convergence of the binding energies compared to their Dunning counterparts. For a number of the tested FPD protocols even def2-TZVPP basis set gives MSDs of $0.5\,kcal\,mol^{-1}$ and smaller.
- Among the considered DFT approximations, we particularly commend PWPB95-D4 double hybrid functional in conjunction with CBS(def2-TZVPP/def2-QZVPP) extrapolation which consistently yields robust binding energies ($MSD = -0.1\,kcal\,mol^{-1}$) and is at least an order of magnitude faster compared to the DLPNO-CCSD(T) based FPD protocols. Good performance with MSD $< 1\,kcal\,mol^{-1}$ is also observed for r^{2}SCAN-D4/D3(BJ) and r^{2}SCAN-3c methods. In contrast, well-established B3LYP and PBE0 functionals complemented with D3(BJ) or D4 dispersion corrections are not recommended for the calculation of the binding energies.

The developed LIMIXCARB_RE12 set can be used for the development, improvement and validation of the new semi-empirical, DFT and approximate WFT methods. Representative low-energy conformers of $Li^+(CC)_2(LC)_2$ clusters and their accurate binding energies are relevant to the theoretical rationalization of the microsolvation processes occurring in the Li-ion batteries.

Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The LIMIXCARB_RE12 database containing PBE0-D3(BJ)/def2-TZVP optimized geometries and reference DLPNO-CCSD(T_1)/CPS(6,7)/CBS(cc-pwCVTZ/cc-pwCVQZ) electronic energies (in E_h) of $Li^+(CC)_2(LC)_2$ clusters and individual organic carbonate molecules and Li^+ ion is available at https://github.com/QuantumChemistryGroup/LIMIXCARB_RE12/tree/main/DATABASE.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Figure S1:** Mean signed deviation (MSD) for the impact of the iterative triples (T_1) correction on the reaction energies. The $\Delta_r E_{IT}$ effect for each reaction is approximated as the difference between DLPNO-CCSD(T_1) and DLPNO-CCSD(T_0) reaction energies obtained with different PNO settings and basis sets. Reference $\Delta_r E_{IT}$ were obtained at AE-DLPNO-CCSD(T)/CPS(6,7)/CBS(cc-pwCVTZ/cc-pwCVQZ) level. Endpoints of each line indicate minimum and maximum deviations among the reactions. The following abbreviations of the basis sets are used: def2-*n*Z = def2-*n*ZVPP (*n* = T, Q). **Figure S2:**. Mean signed deviation (MSD) for the effect of PNO settings on the reaction energies. The $\Delta_r E_{PNO}$ effect for each reaction is calculated as the difference between CPS(5,6)/*TightPNO*/CPS(6,7) and *NormalPNO* DLPNO-CCSD(T_1) reaction energies obtained with different basis sets (or with semi-canonical T_0 triples in case of TZ basis set). Reference $\Delta_r E_{PNO}$ were obtained at AE-DLPNO-CCSD(T)/CPS(6,7)/CBS(cc-pwCVTZ/cc-pwCVQZ) level. Endpoints of each line indicate the minimum and maximum deviations among the reactions. The following abbreviations of the basis sets are used: def2-*n*Z = def2-*n*ZVPP (*n* = T, Q).

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